

**FEDERAL AVIATION ADMINISTRATION
STATEMENT OF OBJECTIVES
for**

**Enhancement of FAALCAn – a Web-Based Life Cycle Assessment (LCA) Tool for
Airports**

1. BACKGROUND

The FAA previously developed guidelines and a framework for conducting LCA studies for airfield infrastructure, with case studies focused on pavement at four major airfields. The scope of guidelines included runways, taxiways, shoulders, aprons, and fences. Follow on work led to the creation of FAALCAn, a web-based LCA tool that quantifies energy use, local aggregate resource consumption, and environmental impacts (air, water) from pavement construction, rehabilitation, maintenance, and end-of-life stages. FAALCAn uses publicly available life cycle inventory data from the Federal LCA Commons, tailored to U.S.-specific materials and equipment, and allows user-customizable inputs for materials, mix designs, construction equipment, transportation modes, and local energy conditions. The web-based platform ensures simultaneous updates for all users, eliminates desktop application compatibility issues, and integrates process libraries from the Federal LCA Commons and OpenLCA.

2. SCOPE OF WORK

This project aims to enhance FAALCAn's functionality by focusing on improved quantification of energy and resource use, integration with FAA tools, and support for local aggregate resource management and supply chain sustainability. Additionally, the project will develop comprehensive user training programs to promote energy and resource conservation as well as cost-effective airfield infrastructure planning.

3. REQUIREMENTS

The Contractor must provide all necessary personnel, technical expertise, operations, administrative support and other resources required to fulfill the tasks outlined in this SOW. Each task will include FAA stakeholder reviews at appropriate milestones.

The Contractor must perform the following tasks:

Task 1: FAALCAn OUTPUTS

Expand FAALCAn's reporting capabilities to include tracking of domestic aggregate resource use and recycled material consumption. This expansion will provide visibility into material sourcing trends by incorporating datasets on regional material availability and recycled content adoption rates, enabling comparisons between conventional and innovative alternatives.

Task 2: FAALCAn INTEGRATION WITH FAARFIELD AND AIRCOST

FAARFIELD is FAA's airport pavement thickness design software and is used to design multiple options (such as concrete pavement, asphalt pavement, etc.). The design information, material sources (such as asphalt plants, aggregate quarries, concrete plants, etc.), and pavement rehabilitation options are input (manually) into Life Cycle Cost Analysis (LCCA) such as AIRCOST to perform LCCA. Pavement design information (from FAARFIELD) and LCCA information is then manually input into FAALCAn to determine energy consumption and material resource usage. The contractor must integrate FAARFIELD, AIRCOST, and FAALCAn so that information (outputs) from FAARFIELD is fed into AIRCOST and resulting information is fed into FAALCAn without user inputting information manually. This integration will result in seamless data (such as material quantities, project schedules, and equipment usage) exchange between FAARFIELD, AIRCOST and FAALCAn reducing manual entry errors and increasing workforce

efficiency. The integration of FAARFIELD, AIRCOST and FAALCAn will allow real-time comparisons of cost, performance, and energy/material resource impacts.

Task 3: DEVELOP A DATABASE OF DOMESTIC SUPPLY CHAINS AND ENERGY SOURCES

Develop a database of domestic supply chains (such as refineries for asphalt binder, aggregate quarries, cement plant, etc.) and their geographic locations, and power sources (electricity suppliers, fuel suppliers, etc.) and implement in FAALCAn. This will enhance FAALCAn to promote energy efficiency and resilient domestic supply chains. These improvements will transform FAALCAn into a more powerful platform for resilient and efficient infrastructure planning, supporting life cycle energy and resource use efficiency in decision-making.

Task 4: FAALCAn TRAINING & WORKSHOPS

Develop a Section 508 compliant training program for FAALCAn aimed at airport authorities and their supporting consultants in the United States. The Contractor must maintain an up-to-date FAALCAn “Help File” that reflects all changes and corrections. The Help File included with the software must remain current. The Contractor must retain the Help File in Microsoft Word/Google Workspace or another platform as directed by COR and use an authoring tool to publish it as a navigable HTML Help File.

4. DELIVERABLES AND SCHEDULE

4.1 Deliverables

4.1.1 Task Plan (to include the tasks listed above) - Task Plans must be submitted to the FAA at the written direction of the FAA Contracting Officer’s Representative (COR). The plans must include a narrative of the objectives of the task, technical approach to be used, a listing of the technical deliverables, and a listing of major milestones with anticipated schedule. The task plan must be returned to the Contractor with review comments within five (5) working days. The Contractor must incorporate the review comments in the task plan before proceeding with work under the task. The Contractor must submit the final task plan to the COR within five (5) working days. The final task plan must be in Microsoft Project or Google Workspace.

4.1.2 Quarterly Reports; 4.1.3 Annual Reports; 4.1.4 Final Report with executive summary slide deck - Draft Technical Reports must be delivered to the COR according to the schedules established in Task Plans. The draft report will be delivered in an electronic file in Microsoft Word or Google Workspace. Preparation of the reports will be in accordance with Technical Report Template (provided by FAA COR). The Government will provide a review of each report within thirty (30) calendar days of delivery. A copy of the final Technical Report after addressing FAA comments will then be delivered within thirty (30) days of receipt of the Government review.

4.1.5 FAALCAn Tool – database, all related software source codes, HTML Help File and related documentation must be provided to the COR.

4.2 Deliverables Schedule

Task	Deliverable Description	Delivery Method	Delivery format	Delivery Schedule
4.1.1	Task Plan	e-mail	Microsoft Project 2000 or later.	15 working days after award of contract

4.1.2	Quarterly Reports	e-mail	Microsoft Word	10 working days after end of every quarter
4.1.3	Annual Reports	e-mail	Microsoft Word	30 working days after end of year
4.1.4	Final report with executive summary slide deck	e-mail/file transfer protocol (FTP)	Microsoft Word/PowerPoint	30 working days before end of the PoP
4.1.5	FAALCAn Tool			30 working days before end of the PoP

5. PERIOD of PERFORMANCE

The Period of Performance (PoP) for this project is 24 months. Any work beyond what is outlined in this SOW requires prior approval from the COR and FAA Contracting Officer (CO). Tasks completed without approval are subject to non-payment.

6. TRAVEL

Travel will be reimbursed on a cost-reimbursable basis as a separate contract line item in accordance with the Federal Travel Regulations.

For planning purposes, plan for three (3) two-day training workshops, one each in Washington, D.C., Chicago, and Seattle. Any travel must be approved by the COR and locations are subject to change.